

Portfolio management of Commodity Trading Advisors with volatility targeting

MARAT MOLYBOGA*

ABSTRACT

I show analytically that a volatility-targeted allocation methodology improves the risk-adjusted performance of portfolios under a broad set of assumptions regarding the serial correlation of returns, and dependence of the expected Sharpe ratio on the level of volatility. I examine the impact of volatility targeting on portfolios of Commodity Trading Advisors within the large-scale simulation framework of Molyboga and L'Ahelec (2016) that accounts for the realistic constraints on institutional investors. I find a consistent and statistically significant improvement in the out-of-sample returns that ranges between 0.53% and 0.80% per annum, on average. The performance enhancement is robust to portfolio size and manager selection, and is implementable inside managed account investments.

*Marat Molyboga is the Chief Risk Officer and Director of Research with Efficient Capital Management and Adjunct Faculty at the Stuart School of Business, Illinois Institute of Technology, 565 W Adams St., Chicago, IL 60661; Email:molyboga@efficient.com. I am grateful to Adam Butler, Adam Duncan, Michael Going, Ernest Jaffarian, Joel Handy, Christophe L'Ahelec, Michael Marcey, Mark Refermat, Chris Solarz, Larry Swedroe and Kevin Townsend for helpful comments.

I. Introduction

Previous research reports that managing volatility improves the performance of risk premia. Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016) show that volatility targeting, or adjusting exposure to target a constant ex-ante volatility, nearly doubles the Sharpe ratio of cross-sectional momentum. Kim, Tse and Wald (2016) report that the abnormal returns of time-series momentum are largely driven by scaling returns by volatility. Moreira and Muir (2017) extend evidence for the benefit of managing the volatility of portfolios to a broad array of risk premia such as market, value, currency carry and betting-against-beta. While the impact of volatility-targeting has been thoroughly examined for risk premia, previous research lacks any guidance for active strategies such as multi-manager portfolios of mutual funds, hedge funds or Commodity Trading Advisors (CTAs). The purpose of this paper is to close that gap by providing theoretical underpinnings for volatility-targeting and investigating empirically whether, and under what conditions, it improves the performance of multi-CTA portfolios.

It is important to distinguish volatility-weighting of portfolio constituents, which closely relates to the concept of diversification across portfolio constituents, from targeting portfolio volatility, which is linked to diversification across the dimension of time. Volatility-weighting is a constant leverage approach highlighted in Maillard, Roncalli and Teiletche (2010) and Hallerbach (2012) that allocates risk equally across portfolio constituents. By contrast, volatility-targeting is an approach to dynamically scaling aggregate portfolio leverage to achieve constant expected portfolio volatility, or allocate risk equally across time rather than portfolio constituents.

It is not obvious whether one should expect an improvement in performance. On the one hand, CTAs may incorporate their expectations regarding the opportunity set and dynamically adjust their risk exposures. If the forecasts are correct and higher levels of volatility are associated with better performance, then targeting the volatility of multi-CTA portfolios may offset the benefits of the forecasts, adversely affecting performance. If, on the other hand, the opportunity set is difficult to forecast, there will be no strong link between volatility and performance. In this case, targeting the volatility of multi-CTA portfolios is more likely to improve performance. The contribution of this paper is three-fold. First, I introduce a simple theoretical model and show that volatility-targeting, under a broad set of assumptions that allows for a limited predictability of

future performance by CTAs, is superior to a constant leverage portfolio approach. Second, I show empirically within the large-scale simulation of Molyboga and L'Ahelec (2016) that volatility-targeting improves the performance of CTA portfolios. Finally, I show that the improvement is robust to portfolio size and manager selection and discuss some of the potential implementation issues and how they can be overcome.

I consider three versions of a theoretical model. The first two versions assume that the expected Sharpe ratio of the risky portfolio is constant. For simplicity, I assume that volatility is known, since it is easy to estimate, as suggested in Merton (1980), and highly persistent, as discussed in Bollerslev and Engle (1993). In the first version, as in Hallerbach (2012), the serial correlation is assumed to be equal to zero whereas, in the second version, it is considered to be a constant non-zero value. I find that in the first two cases volatility targeting is the optimal strategy to maximize the expected Sharpe ratio. This result is intuitive and related to diversification. Since the opportunity set, measured in terms of expected Sharpe ratio, is constant across time, the optimal risk allocation strategy is to maximize diversification across time by taking a constant risk exposure. Moreover, since the risk exposure is proportional to volatility, the optimal allocation, or leverage, is inversely related to volatility.

The third version considers that the expected Sharpe ratio might depend on the level of volatility. Though the optimal strategy depends on the specific parameters that describe the link between volatility and the future expected Sharpe ratio, I find that volatility-targeting is superior to a naive constant leverage approach for a broad set of assumptions that allows for a weak positive relationship between volatility and the expected Sharpe ratio.

Once I establish a theoretical basis for applying volatility targeting, I investigate empirically whether the approach improves the performance of CTA portfolios. I show that the performance of the Barclay CTA Index satisfies the assumptions of the constant Sharpe zero serial-correlation model and, therefore, volatility targeting should prove its performance. However, since the Barclay CTA Index is not investable, I examine the performance impact of targeting volatility of multi-CTA portfolios with a discrete number of CTAs. I consider the two portfolio construction approaches of equal notional and equal volatility-adjusted weightings (described in detail in Appendix C) and evaluate the impact of volatility targeting within the large-scale simulation framework of Molyboga and L'Ahelec (2016) with realistic constraints imposed on a CTA sample of 528 live and 1,113

defunct funds over the 2002-2016 period.

It is important to point out that evaluation of the benchmark approaches themselves is outside the scope of this paper and their optimality heavily depends on cross-sectional assumptions regarding Sharpe ratios, volatilities and correlations. Instead, I focus on determining whether targeting volatility of the benchmark portfolios enhances the performance relative to the naive constant leverage approach applied to the benchmark portfolios. There is a direct link between my empirical tests and the theoretical model once we consider that a benchmark approach is the single asset introduced in the theoretical model section.

I find that volatility targeting improves the out-of-sample net-of-fee Sharpe ratio¹ of portfolios with ten CTAs by approximately 13%-14%, on average. This improvement in the Sharpe ratio represents approximately 63-68 basis points of return per annum for a 15% volatility portfolio. I vary the number of managers in the portfolio between five and twenty to investigate the sensitivity of the performance improvement to the portfolio size. I find that volatility targeting consistently improves performance by 0.53% to 0.80% per annum with greater performance enhancement for larger portfolios.

I further investigate whether volatility targeting is robust to manager selection by measuring the percentage of simulations that benefit from volatility targeting. For the equal notional approach, volatility targeting improves performance in 77.65% of scenarios among portfolios of five managers and grows monotonically to 94.82% for portfolios of twenty managers. For the equal volatility-adjusted allocation, the improvement is observed in 72.37% of scenarios for portfolios of five managers and grows monotonically to 89.54% for portfolios of twenty managers. These findings suggest that the improvement in performance derived from volatility targeting is not sensitive to manager selection.

Finally, I discuss whether investors can capture the benefit of volatility-targeting after accounting for real life constraints such as trading frictions due to monthly rebalancing and portfolio leverage changes. I show that while the frictions are potentially high for fund investors because of a time gap between a redemption from one fund and a subscription into another fund, the frictions can be mitigated if the investor either rebalances among funds of the same hedge fund manager or utilizes managed accounts that employ notional funding and, therefore, are immune to the is-

¹The dataset of CTAs is comprised of net-of-fee returns, as discussed in detail in the Data section.

sue of redemption delays. The rest of the paper is organized as follows: Section 2 introduces a theoretical model; Section 3 describes the data; Section 4 discusses a large-scale simulation and presents empirical results; and Section 5 concludes with how our findings can be broadly applied to multi-manager portfolios of hedge funds and mutual funds.

II. A theoretical model

In this section, I consider a few simple T -period models with a single risky asset and a risk-free asset. The first two models assume that the expected Sharpe ratio of the risky portfolio is constant. For simplicity, I assume that volatility is known, or easy to estimate, as suggested in Merton (1980), and highly persistent, as discussed in Bollerslev and Engle (1993). I denote $\hat{\sigma}_t$ as volatility at time t . The serial correlation is assumed to be equal to zero in the first model and a constant value, ρ , in the second model. The third model considers that the expected Sharpe ratio might depend on the level of volatility.

I denote $\tilde{r}_t$, the nominal log-return of risky asset at time t , and rf_t , the nominal log-return of the risk-free asset at time t . Thus, the excess log return of the risky asset at time t is equal to $r_t = \tilde{r}_t - rf_t$. An expectation of a constant Sharpe ratio, S , implies that

$$\frac{r_t}{\hat{\sigma}_t} = S + \epsilon_t, \quad (1)$$

where $\epsilon_t \sim N(0, 1)$ is a standard normal variable, or, equivalently,

$$r_t = \hat{\sigma}_t(S + \epsilon_t). \quad (2)$$

I assume that at time t the allocation to the risky asset is equal to λ_t . For example, holding a single unit of the risky asset at time t corresponds to $\lambda_t = 1$ and holding two units of the risky asset at time t corresponds to $\lambda_t = 2$. Henceforth, I refer to a portfolio with dynamic λ_t as a dynamic leverage portfolio and to a portfolio with constant $\lambda_t = 1$ as a naive constant leverage portfolio.

Since the return of the leveraged portfolio at time t is equal to $\lambda_t r_t$, the cumulative leveraged return of a portfolio over a horizon of T is equal to $R_T(w) = \lambda_1 r_1 + \dots + \lambda_T r_T = \sum_{t=1}^T \lambda_t r_t$. Given assumption (2) and denoting by $\lambda = (\lambda_1, \dots, \lambda_T)'$, the cumulative excess return of the portfolio is

equal to

$$R_T(\lambda) = \sum_{t=1}^T \lambda_t r_t = \sum_{t=1}^T \lambda_t \hat{\sigma}_t (S + \epsilon_t) = S \sum_{t=1}^T \lambda_t \hat{\sigma}_t + \sum_{t=1}^T \lambda_t \hat{\sigma}_t \epsilon_t. \quad (3)$$

While most allocation methodologies such as risk parity, discussed in detailed in Maillard et al. (2010), consider allocation of risk across assets or portfolio constituents, λ represents allocation of risk across time.

The Sharpe ratio,² SR , of the portfolio is equal to

$$SR(\lambda) = \frac{E(R(\lambda))}{\sqrt{Var(R(\lambda))}} \quad (4)$$

Using expression (3), the expected return of the portfolio is equal to

$$E(R_T(\lambda)) = E\left(S \sum_{t=1}^T \lambda_t \hat{\sigma}_t + \sum_{t=1}^T \lambda_t \hat{\sigma}_t \epsilon_t\right) = S \left(\sum_{t=1}^T \lambda_t \hat{\sigma}_t\right). \quad (5)$$

A. Zero serial-correlation model

The first model assumes that returns are serially uncorrelated (i.e., $cor(\epsilon_t, \epsilon_\tau) = E(\epsilon_t \epsilon_\tau) = 0$ if $t \neq \tau$). Using expression (3), the variance of the leveraged portfolio cumulative excess return is equal to

$$Var(R_T(\lambda)) = Var\left(S \sum_{t=1}^T \lambda_t \hat{\sigma}_t + \sum_{t=1}^T \lambda_t \hat{\sigma}_t \epsilon_t\right) = \sum_{t=1}^T (\lambda_t \hat{\sigma}_t)^2, \quad (6)$$

and, thus, the Sharpe ratio is equal to

$$SR(\lambda) = \frac{E(R(\lambda))}{\sqrt{Var(R(\lambda))}} = S \frac{\sum_{t=1}^T \lambda_t \hat{\sigma}_t}{\sqrt{\sum_{t=1}^T (\lambda_t \hat{\sigma}_t)^2}}. \quad (7)$$

Using the inequality from Appendix A, I get $SR(\lambda) \leq S\sqrt{T}$ and the equality is obtained when λ_t , the leverage value at time t , is inversely related to volatility $\hat{\sigma}_t$ (i.e., $\lambda_1 \hat{\sigma}_1 = \dots = \lambda_T \hat{\sigma}_T$). Thus, in the case of the zero-correlation model, the optimal dynamic leverage values are inversely related to volatilities.

This result is intuitive and related to diversification. Since the opportunity set, measured in

²I ignore the scaling parameter in my calculation because it is going to have the same impact on all allocation strategies.

terms of the expected Sharpe ratio, is constant across time, the optimal risk allocation strategy is to maximize diversification across time by taking a constant risk exposure. Moreover, since the risk exposure is proportional to volatility, the optimal leverage is inversely related to volatility.

B. Constant serial-correlation model

The second model assumes that excess returns have constant correlation of ρ (i.e., $\text{cor}(\epsilon_t, \epsilon_\tau) = E(\epsilon_t \epsilon_\tau) = \rho$ if $t \neq \tau$). Using expression (3), the variance of the leveraged portfolio cumulative excess return is equal to

$$\begin{aligned} \text{Var}(R_T(\lambda)) &= \text{Var} \left(S \sum_{t=1}^T \lambda_t \hat{\sigma}_t + \sum_{t=1}^T \lambda_t \hat{\sigma}_t \epsilon_t \right) = \sum_{t=1}^T (\lambda_t \hat{\sigma}_t)^2 + \rho \sum_{t \neq \tau} \lambda_t \hat{\sigma}_t \lambda_\tau \hat{\sigma}_\tau = \\ &= (1 - \rho) \sum_{t=1}^T (\lambda_t \hat{\sigma}_t)^2 + \rho \left(\sum_{t=1}^T \lambda_t \hat{\sigma}_t \right)^2. \end{aligned} \quad (8)$$

and, thus, the Sharpe ratio is

$$\begin{aligned} SR(\lambda) &= \frac{E(R(\lambda))}{\sqrt{\text{Var}(R(\lambda))}} = S \frac{\sum_{t=1}^T \lambda_t \hat{\sigma}_t}{\sqrt{(1 - \rho) \sum_{t=1}^T (\lambda_t \hat{\sigma}_t)^2 + \rho \left(\sum_{t=1}^T \lambda_t \hat{\sigma}_t \right)^2}} = \\ &= \frac{S}{\sqrt{\rho + (1 - \rho) \frac{\sum_{t=1}^T (\lambda_t \hat{\sigma}_t)^2}{\left(\sum_{t=1}^T \lambda_t \hat{\sigma}_t \right)^2}}}. \end{aligned} \quad (9)$$

Using the inequality from Appendix A, I get $SR(\lambda) \leq \frac{S}{\sqrt{\rho + \frac{1-\rho}{T}}} = \frac{S\sqrt{T}}{\sqrt{1+(T-1)\rho}}$ and the equality is obtained when leverage values λ_t are inversely related to volatility $\hat{\sigma}_t$ (i.e., $\lambda_1 \hat{\sigma}_1 = \dots = \lambda_T \hat{\sigma}_T$). Thus, in the case of the constant-correlation model, optimal leverage values are also inversely related to volatility.

C. Time-variate Sharpe model

The third model assumes a linear relation between the expected Sharpe ratio and volatility where

$$S_t = S_0 + \alpha \hat{\sigma}_t. \quad (10)$$

The slope coefficient, α , can be either positive or negative. I also assume that S_0 is positive.

I further assume that

$$r_t = \hat{\sigma}_t(S_t + \epsilon_t), \quad (11)$$

where $\epsilon_t \sim N(0, 1)$ is a standard normal variable and the returns are serially uncorrelated (i.e., $\text{cor}(\epsilon_t, \epsilon_\tau) = E(\epsilon_t \epsilon_\tau) = 0$ if $t \neq \tau$).

Given assumption (10), the leveraged portfolio cumulative return is equal to

$$R_T(\lambda) = \sum_{t=1}^T \lambda_t r_t = \sum_{t=1}^T \lambda_t \hat{\sigma}_t (S_0 + \alpha \hat{\sigma}_t + \epsilon_t) = S_0 \sum_{t=1}^T \lambda_t \hat{\sigma}_t + \alpha \sum_{t=1}^T \lambda_t \hat{\sigma}_t^2 + \sum_{t=1}^T \lambda_t \hat{\sigma}_t \epsilon_t \quad (12)$$

and the Sharpe ratio is

$$SR(\lambda) = \frac{E(R(\lambda))}{\sqrt{\text{Var}(R(\lambda))}} = \frac{S_0 \sum_{t=1}^T \lambda_t \hat{\sigma}_t + \alpha \sum_{t=1}^T \lambda_t \hat{\sigma}_t^2}{\sqrt{\sum_{t=1}^T (\lambda_t \hat{\sigma}_t)^2}}. \quad (13)$$

Since the optimal solution relies heavily on the assumption of S_0 and α , I compare two approaches: the naive constant leverage approach with the constant allocation to the risky asset, λ^n , such that $\lambda_1^n = \dots = \lambda_T^n = 1$ and an alternative approach of the dynamic leverage with constant volatility targeting, λ^σ , such that $\lambda_1^\sigma \hat{\sigma}_1 = \dots = \lambda_T^\sigma \hat{\sigma}_T = 1$.

Appendix B includes derivations that show that the naive constant leverage approach generally underperforms the alternative approach of the dynamic leverage with volatility targeting, except under a very strict set of conditions that requires α to be positive and the difference of S^+ and S^- , normalized by the average Sharpe ratio $\hat{S}$, to satisfy

$$\frac{S^+ - S^-}{\hat{S}} \geq \frac{2\sigma_\sigma}{\sqrt{\bar{\sigma}^2 + \sigma_\sigma^2 + \bar{\sigma}}}, \quad (14)$$

where $\bar{\sigma} = \frac{1}{T} \sum_{t=1}^T \hat{\sigma}_t$ is the average level of volatility, $\sigma_\sigma = \sqrt{\frac{1}{T} \sum_{t=1}^T \hat{\sigma}_t^2 - \bar{\sigma}^2}$ is the standard deviation of volatility. $\hat{S}$, S^+ and S^- are the values of the Sharpe ratios at the average level of volatility $\bar{\sigma}$ and the levels of volatility at plus one σ_σ and minus one σ_σ , respectively.

Inequality (14) is very restrictive. If the average level of volatility, $\bar{\sigma}$, is equal to 10% and its volatility, σ_σ , is equal to 2%, then the ratio on the right hand side of the inequality is equal to 0.2. Thus, if the average Sharpe ratio, $\hat{S}$, is equal to 0.5, the difference in the Sharpe ratios associated with higher volatility and lower volatility periods has to be greater than 0.1. This

positive difference is very significant in economic terms yet inconsistent with previous empirical findings such as Harvey (2001) which found no strong empirical evidence for a positive association between the stock market variance and expected return and Ang and Liu (2007) which argued that the relation can even be negative under plausible assumptions regarding the dynamics of expected returns and variance.

Therefore, I conclude that the dynamic leverage approach that targets volatility is likely to outperform the naive constant leverage approach even if there is a weak positive relationship between volatility level and expected Sharpe.

III. Data

In this paper, I use the BarclayHedge database, the largest publicly available database of Commodity Trading Advisors, with 528 active and 1,113 defunct funds between December 2002 and December 2016. Joenvaara, Kosowski and Tolonen (2012) present results of a comprehensive study of publicly available databases and report that the BarclayHedge data is of superior quality to the data in the other databases considered. We account for the biases in the data following Blocher and Molyboga (2017). Specifically, I include the graveyard database of defunct funds to account for the survivorship bias and adjust for the backfill/incubation bias by using the first reported date as recommended in Bhardwaj, Gorton and Rowenhorst (2014), Foran et al. (2017) and Jorion and Schwarz (2017). The backfill/incubation bias arises due to the voluntary nature of self-reporting to hedge fund databases.³ Bhardwaj et al. (2014) show that the common adjustment of removing 24 months of returns is inadequate for CTAs because the remaining bias is still more than 1% per annum in a value-weighted index and almost 3% in an equally-weighted index. Since the first reported date is available in the BarclayHedge database starting in December 2002, our dataset is limited to December 2002 through December 2016. We include defunct managers to account for the survivorship bias. The selection bias that is driven by funds choosing not to report to public databases is the most difficult to avoid, but it's likely to be insignificant, as reported in Edelman, Fung and

³Funds often go through an incubation period during which they build a track record using proprietary capital. Fund managers start reporting to a CTA database to raise external capital only if the track record is attractive and they often "backfill" the returns generated prior to inclusion in the database. The incubation/backfill bias arises since funds with poor performance are unlikely to report their performance to the database.

Hsieh (2013).⁴ We use the liquidation estimate of 1% as suggested in Ackermann, McEnally and Ravenscraft (1999). Since CTA investments are often chosen for their diversification benefit during crisis periods, as summarized in Molyboga and L'Ahelec (2016), I exclude fund managers that are most susceptible to market crisis such as those that predominantly rely on options or arbitrage.⁵ I use monthly values of the 3-month T-bills secondary market rate with ticker "TB3MS" from FRED as a proxy of the risk-free rate. I use the Barclay CTA Index from the BarclayHedge database as a proxy of aggregate performance of CTAs.

Table I reports average summary statistics for the defunct managers, live managers and all managers in the CTA sample. The statistics include mean monthly excess return, median monthly excess return, monthly standard deviation, skewness, excess kurtosis, track record length, total number of managers, average number of managers, number of observations and average AUM in millions of US dollars. As expected, live managers tend to perform better, stay in business longer and attract more assets than defunct managers. The mean monthly excess return of live managers is 0.28%, exceeding the 0.06% mean return of the defunct managers. The corresponding t-statistic of the difference of the means is equal to 4.14, a value which is significant at the 99% level of confidence. The average track record length of live and defunct managers is 72 and 48 months, respectively. The average AUM of live and defunct managers is U.S. \$797 million and U.S. \$162 million, respectively. The high mortality rate among CTA funds is indicated by the large share of defunct funds in the database. Out of 1,641 funds in the sample, 1,113 funds are defunct and only 528 are live.

IV. Empirical Analysis

In this section, I describe the large-scale simulation framework of Molyboga and L'Ahelec (2016) that was applied to the portfolios in the study and present the empirical results. Prior to running empirical analysis for CTA portfolios, I test whether the assumptions imposed in the previous section are valid at the aggregate level by examining the Barclay CTA Index. I estimate volatility

⁴There are two types of funds that choose not to report in public databases: poorly performing funds that cannot attract new investors based on their track record and successful funds that do not rely on the commercial databases to raise assets. Edelman et al. (2013) show that the two sources of non-reporting bias cancel each other out.

⁵I use the BarclayHedge classification field to remove funds with three classifications: Option Strategies, Stock Index/Option Strategies, and Arbitrage. I repeat the analysis including all manager classifications and find that the key results are very similar.

using sample standard deviation of excess returns of the Barclay CTA Index over the previous 36 month and is used as an ex-anti volatility prediction for the following month. Each standardized return is calculated by dividing excess return by the corresponding ex-anti volatility. It is worth noting that standardized returns are equal to Sharpe ratios up to a constant scaling parameter.

I apply Ljung-Box test with up to 12 lags to the standardized returns and find no evidence of serial correlation. The p-values range between 19% for the lag of 4 and 80% for the lag of 12. Therefore, I conclude that the standardized returns, or the Sharpe ratios, are serially uncorrelated.

Next I examine a relationship between the time-series of ex-anti volatility and standardized returns. Figure 1 displays a scatter plot of the time-series ex-anti volatility and standardized returns for the Barclay CTA index for the period between January 2006 and December 2016. There is no obvious linear relationship between ex-anti volatilities and standardized returns. The correlation coefficient is equal to -0.06, which is statistically insignificant from zero. Therefore, I conclude that the performance of the Barclay CTA Index satisfies the assumptions of the constant Sharpe zero serial-correlation model considered in Section II and, therefore, volatility targeting should improve performance.

Since the Barclay CTA Index is not investable, I examine the impact of targeting volatility on the performance of multi-CTA portfolios with a discrete number of managers by comparing the naive constant leverage approach against a dynamic leverage approach with volatility targeting. I consider two benchmark portfolio construction approaches: the 1/N naive diversification approach (designated EN for equal notional) highlighted by DeMiquel, Garlappi and Uppal (2009) and the equal volatility-adjusted approach (EVA) introduced by Hallerbach (2012).

It is important to point out that evaluation of the benchmark approaches themselves is outside the scope of this paper and their optimality heavily depends on cross-sectional assumptions regarding Sharpe ratios, volatilities and correlations. Instead, I focus on determining whether targeting volatility of the benchmark portfolios enhances the performance relative to the naive constant leverage approach applied to the benchmark portfolios. There is a direct link between my empirical tests and the theoretical model once we consider that a benchmark approach is the single asset introduced in the theoretical models of Section II.

I estimate portfolio volatility using a Ledoit and Wolf (2004) shrinkage estimator applied to an exponentially-weighted sample covariance matrix of the portfolio constituents' returns. Appendix

C provides a detailed definition of EN, EVA and two distinct approaches to targeting volatility: VTEN, the volatility targeting version of EN, and VTEVA, the volatility targeting version of EVA. Both volatility-targeting approaches target volatility of 12% per annum. It is important to note that since volatility is not known, forecasts based on realized volatility are used. While the theoretical model uses excess log-returns, the simulation relies on nominal excess returns, which has implications for the calculation of Sharpe ratios.

A. A large-scale simulation framework

I rely on the large scale simulation framework of Molyboga and L'Ahelec (2016) with 10,000 simulations for the out-of-sample period between January 2006 and December 2016. The simulation framework is designed to evaluate portfolio construction approaches subject to the realistic constraints of institutional investors. The first allocation decision is made at the end of December 2005. Because of the delay in CTA reporting, documented in Molyboga, Baek and Bilson (2017), an institutional investor can rely on information only through November 2005. Following Molyboga and L'Ahelec (2016) and Molyboga and L'Ahelec (2017), the simulation accounts for two standard constraints that institutional investors face by considering only funds with at least 36 months of returns and excluding all funds in the bottom quintile of AUM among the funds considered. Then the simulation randomly selects ten funds from the available pool of funds and applies four portfolio construction approaches: EN, EVA, VTEN and VTEVA. For each approach, monthly returns are recorded for January 2006 using the liquidation bias adjustment, if needed. At the end of January 2006, the available fund pool is updated to comply with the requirements of institutional investors noted above, and all funds from the original portfolio that no longer meet the requirements are randomly replaced with funds from the new pool. Each portfolio is rebalanced using one of the four methodologies and the process is repeated for the entire out-of-sample period, January 2006-December 2016. A single simulation run produces four time-series of out-of-sample performance, one for each portfolio construction approach. Distributions of out-of-sample returns are constructed based on the results of 10,000 simulations. I repeat the analysis by varying the number of managers in the portfolio between five and twenty for robustness.

Table II reports the average AUM threshold level for each year and the average number of active funds that have at least 36 months of returns and meet that threshold. The variation in the AUM

threshold and the number of active funds reflects industry dynamics.

B. Empirical results

In this section, I present empirical out-of-sample results obtained within the large-scale simulation framework. I show that volatility-targeting improves the Sharpe ratio for each of the two portfolio construction approaches considered. First, I compare the distributions of Sharpe ratios generated for portfolios of ten managers using distribution quartiles. Second, I focus on the means and medians of distributions but vary the number of managers between five and twenty.

B.1. Ten manager portfolios

The large-scale simulation framework produces distributions of out-of-sample performance. Molyboga, Baek and Bilson (2017) introduce a methodology for comparing distributions that relies on utility functions and stochastic dominance. In this paper, I primarily focus on means and medians but also consider distribution quantiles based on this methodology. Table III reports the mean, standard deviation and quantiles of the Sharpe distribution, including the minimum, first quartile, median, third quartile and maximum values. The distributions of Sharpe ratios are generated for the four portfolio construction approaches within the large scale simulation framework. The improvement due to volatility targeting is striking. The five quantiles of the volatility-targeted equal notional (VTEN) approach are consistently higher than those of the standard EN approach. The median Sharpe ratio of 0.368 for the VTEN method is approximately 13% higher than the median Sharpe ratio of 0.326 for the standard EN approach. This difference is significant at the 99% confidence level. Statistical inference utilizes the bootstrap methodology of Molyboga and L'Ahelec (2016). Since the simulations are not independent, a bootstrap approach is required to eliminate the bias inherent in standard tests, as discussed in detail in Molyboga, Baek and Bilson (2017). This improvement in the Sharpe ratio is striking because it translates to 0.63% per annum for a portfolio that targets annualized volatility of 15%. The improvement in the median Sharpe ratio due to volatility targeting is even higher for EVA with a relative improvement of approximately 14% from 0.328 to 0.374. This improvement is statistically significant at the 99% significance level and is equivalent to approximately 0.68% per annum of return enhancement for a 15% volatility portfolio.

Distributions of Sharpe ratios can be visualized using box and whiskers plots. The breadth of each distribution represents the role of chance due to manager selection, which has been highlighted in Molyboga and L'Ahelec (2016). Figure 2 shows the distributions of the Sharpe ratios of the hypothetical portfolios, generated within the large-scale simulation framework for the out-of-sample period of January 2006-December 2016. The chart shows that volatility targeting consistently improves the Sharpe ratios for both EN and EVA.

B.2. Robustness to portfolio size

In the previous section, I have established that volatility targeting consistently improves the distribution of the Sharpe ratios for both EN and EVA. In this section, I examine whether the improvement is robust to portfolio size by varying the number of managers in the portfolios between five and twenty.

Table IV reports the mean and median improvements in the annualized return due to volatility targeting. The number of managers ranges between five and twenty. The distributions of Sharpe ratios are generated for the four portfolio construction approaches within the large-scale simulation framework and the analysis of return improvement is based on a 15% target volatility. The improvement in return due to volatility targeting is statistically significant at the 99% level in each case and grows monotonically with the number of managers in the portfolio. This pattern is consistent with the findings of Blocher, Jiang and Molyboga (2016) who present evidence that the Barclay CTA index follows a random walk and, thus, the opportunity set is difficult to predict for portfolios with a large number of CTAs.

The return improvement ranges between 0.53% to 0.69% per annum for EN and between 0.59% and 0.80% for EVA, on average. The median return is 0.55% to 0.70% higher for EN and by 0.63% to 0.79% for EVA. Thus, I conclude that the average improvement due to volatility targeting is significant both in economic and statistical terms, and robust to the number of managers in the portfolio.

B.3. Robustness to manager selection

In the previous section, I have shown that targeting the volatility of portfolios improves the out-of-sample performance, on average. In this section, I look beyond average performance improvement

and investigate the pervasiveness of the improvement across the 10,000 simulations.

Table V reports the percentage of scenarios with improvement in performance due to volatility targeting. The number of managers ranges between five and twenty. I find that the improvement in performance is robust. For the equal notional approach, volatility targeting improves performance in 77.65% of scenarios for portfolios with five managers, growing monotonically to 94.82% of scenarios for portfolios of twenty managers. For the equal volatility-adjusted allocation, the improvement is observed in 72.37% of scenarios for portfolios of five managers and grows monotonically to 89.54% of scenarios for portfolios with twenty managers. These findings suggest that the improvement in performance derived from volatility targeting is not sensitive to manager selection.

C. Implications for investors

In this section, I discuss the implications of the theoretical model predictions and empirical findings for institutional investors. In particular, the most important question to be addressed is whether investors can capture the potential benefit of volatility targeting after accounting for real life constraints.

The answer to that question is not simple. On the one hand, the methodology of Molyboga and L'Ahelec (2016) accounts for several important constraints that institutional investors face such as size and track record length while attempting to produce implementable results by carefully adjusting for data biases and eliminating the look ahead bias that is present in most hedge fund studies. Therefore, the out-of-sample results produced within the framework are relevant.

On the other hand, the monthly rebalancing and monthly leverage changes that are required for the implementation of volatility targeting are not readily available to fund investors because of a time gap between a redemption from one fund and a subscription into another fund. For example, if an investor de-allocates U.S. \$1,000,000 from fund A, he needs to submit a redemption notice several days prior to the redemption date and it may take several more days to receive cash, fill out a subscription notice and wire the cash to fund B. Thus, during each rebalance the investor is not going to be fully invested during several days or weeks depending on the liquidity provisions of the funds.

The rebalancing delay issue can be mitigated if the investor either rebalances among funds of the same hedge fund manager or utilizes managed accounts that employ notional funding and,

therefore, are immune to the issue of redemption delays. Thus, an investor in several hedge fund managers can fully benefit from the quantitative edge embedded in volatility targeting portfolios only if he uses managed accounts.

V. Concluding remarks

In this paper, I show analytically that a volatility-targeted allocation methodology improves the risk-adjusted performance of portfolios under a broad set of assumptions regarding the serial correlation of returns, the variability of volatility and the dependence of the expected Sharpe ratio on the level of volatility.

I empirically examine the impact of volatility targeting on portfolios of Commodity Trading Advisors within a large-scale simulation framework of Molyboga and L'Ahelec (2016) that accounts for realistic constraints of institutional investors. I find a consistent improvement in the out-of-sample returns that is statistically significant and ranges between 0.53% to 0.80% per annum, on average. The performance enhancement is robust to portfolio size and manager selection, and is implementable with managed account investments.

The theoretical framework of this research and the empirical methodology can be broadly applied to determine whether volatility-targeting can improve the performance of multi-manager portfolios of hedge funds and mutual funds.

Tables and Figures

Table I

Summary statistics of CTAs. This table reports average summary statistics for the defunct managers, live managers and all managers in the CTA sample. The statistics include mean monthly excess return, median monthly excess return, monthly standard deviation, skewness, excess kurtosis, track record length, total number of managers, average number of managers, number of observations and average AUM in millions of U.S. dollars.

	Defunct	Live	All
Mean	0.06%	0.28%	0.13%
Median	0.00%	0.23%	0.07%
StDev	3.71%	4.12%	3.84%
Skewness	0.12	0.18	0.14
Excess kurtosis	1.08	1.06	1.07
Auto-correlation	-0.01	-0.05	-0.02
Track record length	48	72	56
Total number of managers	1,113	528	1,641
Average number of managers	317	225	542
Number of observations	53,591	38,029	91,620
Average AUM (millions US\$)	162	797	367

Table II

Annual statistics of CTAs. This table reports the 20th percentile of the average AUM by year and the number of funds with at least 36 months of returns used in the study.

Date	AUM	CTAs
2006	11,734,900	148
2007	10,670,625	156
2008	12,172,042	169
2009	12,482,167	198
2010	14,843,900	221
2011	14,164,425	242
2012	12,383,375	249
2013	10,880,767	244
2014	10,655,258	223
2015	10,159,425	212
2016	9,908,667	214

Table III

Statistics of Sharpe distributions. This table reports the mean, the standard deviation and the minimum, first quartile, median, third quartile and maximum values of the distribution. The distributions of the Sharpe ratios are generated for the four portfolio construction approaches within the large-scale simulation framework. The out-of-sample period is from January 2006 to December 2016. The p-values of the differences in the means of Sharpe ratios of VTEN and EN as well as VTEVA and EVA are calculated using the bootstrap approach of Molyboga and L'Ahelec (2016). *, **, and *** indicate significance at 10%, 5% and 1% levels, respectively.

	EN	EVA	VTEN	VTEVA
Mean	0.323	0.325	0.364***	0.372***
StDev	0.101	0.126	0.109	0.128
Max	0.770	0.875	0.839	0.932
Third Quartile	0.390	0.407	0.437	0.456
Median	0.326	0.328	0.368	0.374
First Quartile	0.258	0.243	0.294	0.290
Min	-0.132	-0.201	-0.096	-0.153

Table IV

Average improvement due to volatility targeting. This table reports the mean and median improvement in the annualized return due to volatility targeting. The number of managers ranges between five and twenty. The distributions of the Sharpe ratios are generated for the four portfolio construction approaches within the large-scale simulation framework and return improvement is based on the 15% target volatility. The out-of-sample period is from January 2006 to December 2016. The p-values are calculated using the bootstrap approach of Molyboga and L'Ahelec (2016). *, **, and *** indicate significance at 10%, 5% and 1% levels, respectively.

	Mean		Median	
	EN	EVA	EN	EVA
5	0.53%***	0.59%***	0.55%***	0.63%***
6	0.54%***	0.61%***	0.58%***	0.61%***
7	0.57%***	0.65%***	0.59%***	0.65%***
8	0.59%***	0.67%***	0.59%***	0.69%***
9	0.60%***	0.69%***	0.59%***	0.71%***
10	0.62%***	0.71%***	0.63%***	0.68%***
11	0.62%***	0.72%***	0.63%***	0.69%***
12	0.63%***	0.72%***	0.65%***	0.74%***
13	0.64%***	0.74%***	0.65%***	0.74%***
14	0.65%***	0.75%***	0.66%***	0.73%***
15	0.66%***	0.76%***	0.67%***	0.73%***
16	0.66%***	0.75%***	0.68%***	0.73%***
17	0.67%***	0.77%***	0.67%***	0.76%***
18	0.68%***	0.78%***	0.68%***	0.78%***
19	0.69%***	0.79%***	0.70%***	0.78%***
20	0.69%***	0.80%***	0.70%***	0.79%***

Table V

Percentage of scenarios with improvement due to volatility targeting. This table reports the percentage of scenarios with improvement in performance due to volatility targeting. The number of managers ranges between five and twenty. The distributions of the Sharpe ratios are generated for the four portfolio construction approaches within the large-scale simulation framework and return improvement is based on the 15% target volatility. The out-of-sample period is from January 2006 to December 2016.

	EN	EVA
5	77.64%	72.37%
6	79.01%	74.05%
7	80.99%	75.28%
8	83.43%	77.66%
9	84.58%	79.62%
10	86.35%	80.51%
11	86.70%	81.24%
12	88.11%	82.57%
13	88.91%	83.20%
14	90.39%	84.59%
15	90.93%	85.50%
16	91.36%	85.92%
17	92.98%	87.29%
18	93.47%	88.30%
19	93.92%	88.75%
20	94.82%	89.54%

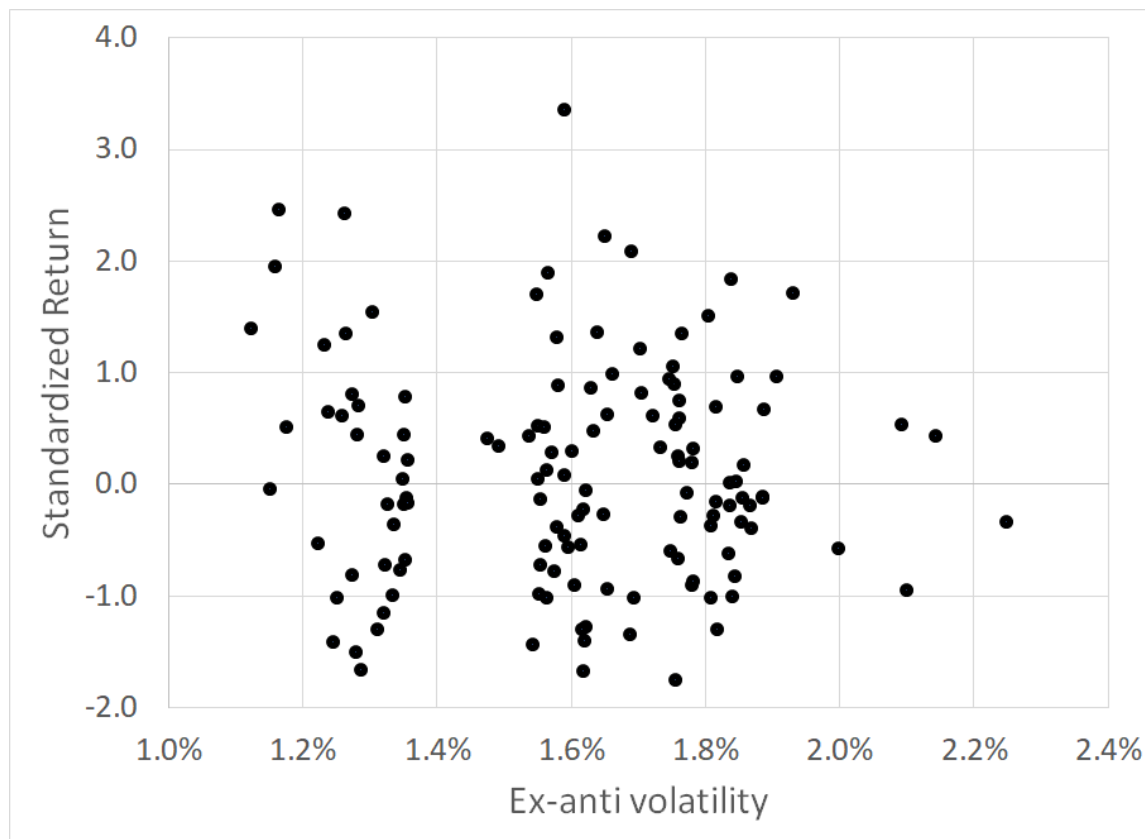


Figure 1. Scatter plot of ex-anti volatility and standardized returns of the Barclay CTA Index. This figure displays a scatter plot of the time-series of ex-anti volatility and standardized returns for the Barclay CTA index for the period between January 2006 and December 2016. Volatility is estimated using sample standard deviation of excess returns of the Barclay CTA Index over the previous 36 month and is used as an ex-anti volatility prediction for the following month. Each standardized return is calculated by dividing excess return by the corresponding ex-anti volatility.

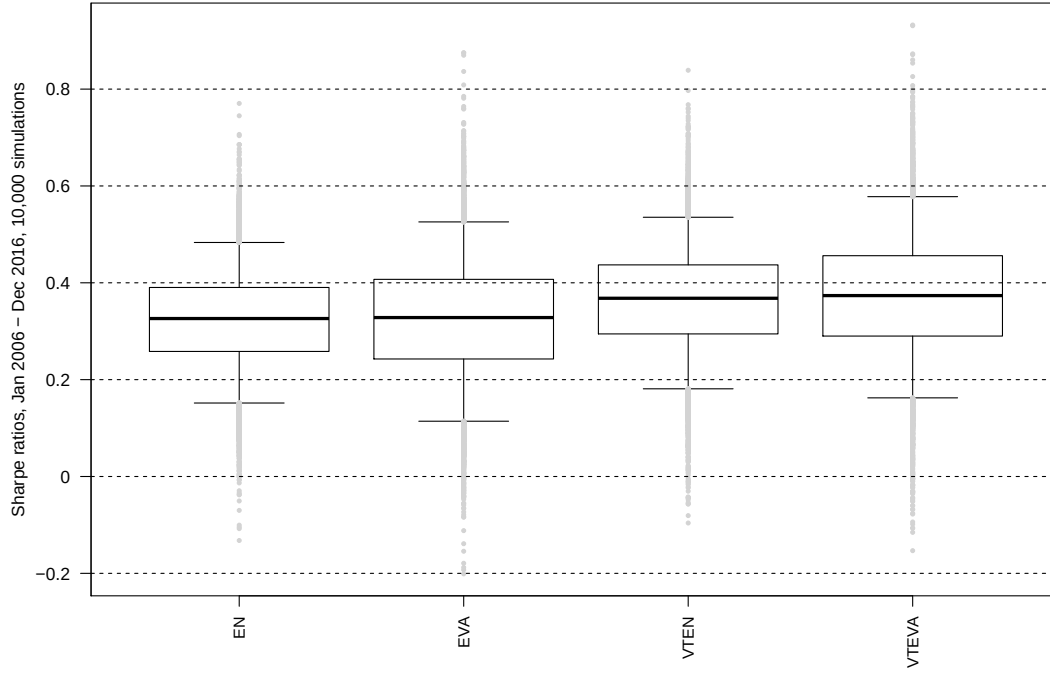


Figure 2. Out-of-sample Sharpe ratios of hypothetical portfolios. This figure displays the distributions of the performance of the hypothetical portfolios, generated within the large-scale simulation framework for the out-of-sample period between January 2006 and December 2016.

Appendices

The appendix includes key derivations and a description of the portfolio construction approaches used in the study.

A. Auxiliary inequality

Proposition 1. Consider a time-series u_t with t ranging between 1 and T .

$$\frac{\sum_{t=1}^T u_t}{\sqrt{\sum_{t=1}^T u_t^2}} \leq \sqrt{T} \quad (15)$$

and the equality is obtained for $u_1 = \dots = u_T$.

Proof: Introduce another time-series v_t . The Cauchy-Schwartz inequality suggests that

$$\left(\sum_{t=1}^T u_t v_t \right)^2 \leq \left(\sum_{t=1}^T u_t^2 \right) \left(\sum_{t=1}^T v_t^2 \right). \quad (16)$$

Since the inequality works for any time-series v_t , it should work for a particularly convenient time-series with all values equal to 1 (e.g., $v_t = 1$ for any t).

$$\left(\sum_{t=1}^T u_t \right)^2 = \left(\sum_{t=1}^T u_t 1 \right)^2 \leq \left(\sum_{t=1}^T u_t^2 \right) \left(\sum_{t=1}^T 1^2 \right) = T \left(\sum_{t=1}^T u_t^2 \right) \quad (17)$$

Re-arranging the terms and taking square roots from both sides produces the expression of proposition 1 and the equality condition is inherited from the Cauchy-Schwartz inequality.

B. Comparison of the naive constant leverage approach and the dynamic leverage approach with volatility targeting

In this section, I compare the Sharpe ratios of two approaches: the naive approach with constant leverage λ^n applied to the risky asset such that $\lambda_1^n = \dots = \lambda_T^n = 1$ and an alternative approach λ^σ such that $\lambda_1^\sigma \sigma_1 = \dots = \lambda_T^\sigma \sigma_T = 1$.

The Sharpe ratio of the naive approach λ^n :

$$SR(\lambda^n) = \frac{S_0 \sum_{t=1}^T \sigma_t + \alpha \sum_{t=1}^T \sigma_t^2}{\sqrt{\sum_{t=1}^T \sigma_t^2}}. \quad (18)$$

The Sharpe ratio of the dynamic approach with volatility targeting λ^σ :

$$SR(\lambda^\sigma) = \frac{E(R(\lambda))}{\sqrt{Var(R(\lambda))}} = \frac{S_0 T + \alpha \sum_{t=1}^T \sigma_t}{\sqrt{T}}. \quad (19)$$

The Sharpe ratio of the naive approach is going to be greater than the Sharpe ratio of the

alternative approach (i.e., $SR(\lambda^n) > SR(\lambda^\sigma)$) iff (if and only if):

$$\alpha > \frac{S_0 \sqrt{T} \left(\sqrt{T} - \frac{\sum_{t=1}^T \sigma_t}{\sqrt{\sum_{t=1}^T \sigma_t^2}} \right)}{\sqrt{T \sum_{t=1}^T \sigma_t^2} - \sum_{t=1}^T \sigma_t} = \frac{S_0 \sqrt{T}}{\sqrt{\sum_{t=1}^T \sigma_t^2}} = \frac{S_0}{\sqrt{\frac{1}{T} \sum_{t=1}^T \sigma_t^2}} > 0. \quad (20)$$

Since S_0 is the intercept of equation (10), it is difficult to interpret. Thus, $\hat{S}$, the Sharpe ratio at the average volatility level $\bar{\sigma} = \frac{1}{T} \sum_{t=1}^T \sigma_t$, is a more meaningful measure. Since $\hat{S} = S_0 + \alpha \bar{\sigma}$ and assuming that $\alpha = \frac{S_0}{\sqrt{\frac{1}{T} \sum_{t=1}^T \sigma_t^2}}$, I get

$$\hat{S} = S_0 + \alpha \bar{\sigma} = S_0 \left(1 + \frac{\bar{\sigma}}{\sqrt{\frac{1}{T} \sum_{t=1}^T \sigma_t^2}} \right) = S_0 \left(1 + \frac{\bar{\sigma}}{\sqrt{\bar{\sigma}^2 + \sigma_\sigma^2}} \right), \quad (21)$$

where $\sigma_\sigma^2 = \frac{1}{T} \sum_{t=1}^T \sigma_t^2 - \bar{\sigma}^2$ represents the variance of σ_t and σ_σ represents the standard deviation of σ_t .

Denote

$$S^+ = S_0 + \alpha(\bar{\sigma} + \sigma_\sigma) = S_0 \left(1 + \frac{\bar{\sigma} + \sigma_\sigma}{\sqrt{\bar{\sigma}^2 + \sigma_\sigma^2}} \right) \quad (22)$$

and

$$S^- = S_0 + \alpha(\bar{\sigma} - \sigma_\sigma) = S_0 \left(1 + \frac{\bar{\sigma} - \sigma_\sigma}{\sqrt{\bar{\sigma}^2 + \sigma_\sigma^2}} \right) \quad (23)$$

as the values of the Sharpe ratios at the levels of volatility of plus one σ_σ and minus one σ_σ , respectively, assuming $\alpha = \frac{S_0}{\sqrt{\frac{1}{T} \sum_{t=1}^T \sigma_t^2}}$.

In order to interpret the economic magnitude of α , I calculate the difference between S^+ and S^- , normalized by the average Sharpe ratio $\hat{S}$:

$$\frac{S^+ - S^-}{\hat{S}} = \frac{2\sigma_\sigma}{\sqrt{\bar{\sigma}^2 + \sigma_\sigma^2} + \bar{\sigma}}. \quad (24)$$

C. Portfolio construction approaches

In this study, I consider two benchmark approaches. For each approach, I use its standard implementation with fixed exposure and an alternative implementation with a constant volatility target.

The two benchmark approaches are EN and EVA:

- The EN allocation is a simple equal notional weight, naive diversification $1/N$ approach:

$$w_i = \frac{1}{N},$$

where N is the number of funds in the portfolio and w_i is the portfolio weight of fund i . By construction, the total notional exposure is fixed since $\sum_{i=1}^N w_i = \sum_{i=1}^N \frac{1}{N} = 1$.

- The EVA approach is based on an equal-volatility methodology with the volatility of each fund, $\hat{\sigma}_i$, estimated using the sample standard deviation of the fund's in-sample returns:

$$w_i = \frac{\frac{1}{\hat{\sigma}_i}}{\sum_{j=1}^N \frac{1}{\hat{\sigma}_j}}.$$

The total notional exposure is fixed since $\sum_{i=1}^N w_i = \sum_{i=1}^N \frac{\frac{1}{\hat{\sigma}_i}}{\sum_{j=1}^N \frac{1}{\hat{\sigma}_j}} = 1$.

For each approach, I introduce a the dynamic volatility-targeted allocation as follows. Denote Σ as the covariance matrix estimated with a Ledoit and Wolf (2004) shrinkage estimator. Given portfolio weights w , the expected portfolio volatility is equal to $\sqrt{w'\Sigma w}$. Thus, to accomplish a target volatility of σ_0 , I apply leverage of $\lambda = \frac{\sigma_0}{\sqrt{w'\Sigma w}}$ to the original portfolio weights, w , as follows:

$$w^{VT} = \lambda w = \frac{\sigma_0}{\sqrt{w'\Sigma w}} w.$$

Thus, the expected portfolio volatility given the portfolio weights, w^{VT} , is equal to

$$\sqrt{\frac{\sigma_0}{\sqrt{w'\Sigma w}} w' \Sigma \frac{\sigma_0}{\sqrt{w'\Sigma w}} w} = \sigma_0 \sqrt{\frac{w' \Sigma w}{\sqrt{w' \Sigma w} \sqrt{w' \Sigma w}}} = \sigma_0.$$

I denote VTEN to be the volatility-targeted version of EN and VTEVA as the volatility-targeted version of EVA.

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